



Review

The effect of graphic organizers on cultivating higher-order thinking: Evidence from a meta-analysis

Hao Lei^a, Yuxin Ding^{a,*} , Ming Ming Chiu^{b,c}

^a Institute of Curriculum and Instruction, Faculty of Education, East China Normal University, Shanghai, China

^b Special Education and Counseling Department, The Education University of Hong Kong, Hong Kong, China

^c Analytics\Assessment Research Centre, The Education University of Hong Kong, Hong Kong, China

ARTICLE INFO

Keywords:

Meta-analysis

Graphic organizer

Higher-order thinking

ABSTRACT

Graphic organizers can pick out and organize key information to release students' cognitive effort from their lower-order thinking for their higher-order thinking (dispositions, cognitive processes, skills), but studies showed mixed results. Hence, this meta-analysis of 4824 students in 66 empirical studies with 72 effect sizes determine the overall effect size of graphic organizers on higher order thinking and moderators that explain the different results. Use of graphic organizers showed an overall positive link with higher order thinking ($g = 0.778$). The effect size is stronger (a) with mind maps, tapering for argument maps and concept maps; (b) in secondary school, tapering for primary school and university; (c) for cognitive processes and dispositions than outcome skills; (d) for divergent thinking than convergent thinking; (e) in countries with higher GDP per capita, and (f) in later publication years. Gender, academic subject and study duration showed no moderation effects. These results inform teachers' judicious use of graphic organizers to boost their students' higher order thinking.

1. Introduction

In our information-rich modern world, education has shifted from understanding it, recalling it, and applying it (*lower-order thinking*, LOT) to analyzing it, evaluating it, and creating it (*higher-order thinking* [HOT]; Bloom, 1956; Zohar and Ben David, 2009). Students with better higher-order thinking (domain general) than others often show more learning motivation (Purnama & Nurdianingsih, 2019), understand ideas more deeply (Thamrin & Widodo, 2019), show greater autonomy and initiative (Smith & Darvas, 2017), and have better learning outcomes in school (e.g. Tanujaya et al., 2017).

Hence, visual tools that pick out key information and spatially organize it (e.g., charts, diagrams, maps; *graphic organizers*, GO) might release students' cognitive effort from their lower-order thinking to their higher-order thinking (Eppler & Burkhard, 2008). However, studies showed mixed results. Graphic organizers boost students' higher-order thinking in some studies (e.g., Achor et al., 2024; Min et al., 2023) but not others (Carwie, 2009; Su & Long, 2021). Differences in participants, settings, or study designs might explain these results. Hence, our meta-analysis of 4824 students in 66 studies (a) determines the overall relation between graphic organizer and higher-order thinking and (b) identifies moderators that account for different results across studies. Understanding graphic organizer's effect on higher-order thinking and factors that boost its effectiveness can help teachers and their students navigate the information avalanche from generative artificial intelligence (e.g., large language models like ChatGPT; Kelly et al., 2023).

* Corresponding author.

E-mail address: 51274102013@stu.ecnu.edu.cn (Y. Ding).

2. Conceptual framework

2.1. Higher-order thinking

Higher-order thinking is defined as comprising dispositions and cognitive processes that yield the skills of analysis, evaluation, inference, deductive reasoning, and inductive reasoning (Hwang et al., 2021). Higher-order thinking dispositions drive higher-order thinking cognitive processes to yield its outcomes/skills. Consider this example. Ana saw dandelions growing in poisoned soil near factories, and was curious about how they thrived. Her open mind led her to many sources, including journal articles and a professor studying how plants remove contaminants. She read about plants that absorbed metals from poisoned earth. These concepts were hard to grasp, but she persisted. Unsatisfied with abstract ideas, she systematically tested them by examining dandelions in clean and dirty soil. The professor generously helped her use special equipment to find that dandelions in polluted soil had more heavy metals and more intricate roots. She assessed these complicated roots to be relevant rather than coincidence. Synthesizing this information about heavy metals and intricate roots, she creatively inferred that genetic mutations gave dandelions such roots, which helped them survive the poisoned soil. Reflecting on her results, she considered other possible causes, like soil pH and water availability, showing her critical thinking and creativity.

Higher-order thinking dispositions are orientations and attitudes, such as Ana's curiosity, open-mindedness, perseverance, and systematicity (Lim, 2017). Her curiosity engages her to ask questions (how do dandelions grow in poisoned soil?), dig for answers, find new ideas, and explore them (Rauscher & Badenhorst, 2021). Her open-mindedness let her see many sides of a story (heavy metals, intricate roots, soil pH) and adapt to such new information (Arviani et al., 2023). Her perseverance keeps her going through difficult tasks and setbacks (e.g., difficult concepts) to eventually master skills (Galita & Bonta, 2023). Her systematicity pushes her to organize her thoughts (dandelions, soil, growth), break down problems into simpler parts they can handle (clean vs. dirty soil), and plan and solve problems (compare dandelions in clean vs. dirty soil; Rauscher & Badenhorst, 2021). Hence, her curiosity and open-mindedness opened doors to new ideas, while her perseverance and systematicity pushed her beyond her initial efforts.

These dispositions motivated Ana's higher-order thinking cognitive processes of analysis, evaluation, synthesis, and inference (Thamrin & Widodo, 2019). She examined dandelions to understand their parts (heavy metals, intricate roots; *analysis*). Then, she assessed the relevance of each part to solving her problem (relevance of roots; *evaluation*). Next, she fit relevant parts together (relate heavy metals to intricate roots; *synthesis*). Based on her organized information, she drew conclusions (genetic mutation aided survival; *inference*).

These higher-order thinking cognitive processes helped her achieve her higher-order thinking outcomes of critiques/critical thinking, creativity, or problem solving (Tikhonova & Kudinova, 2015). Analysis and evaluation especially aid critical thinking (why intricate roots?), and synthesis and inference particularly facilitate creativity (genetic mutation can aid survival). Nevertheless, any of these cognitive processes might support critical thinking, creativity or problem solving (as in Ana's example).

2.2. Graphic organizer

Graphic organizer is defined as a visual tool that pick out key information and organize it (e.g., charts, diagrams, maps; Eppler & Burkhard, 2008). We can organize information via a *spatial metaphor*: time is space; ideas are objects at locations; ideas with shared attributes look similar (color, shape, size, etc.); related ideas are nearby objects; causation is moving from one location to another, and so on (Lakoff & Johnson, 2024). Applying this spatial metaphor, graphic organizers pick out key ideas, organize them and their relations, and store them on a surface (e.g., paper, screen), which reduces the cognitive load (Mayer, 2002) of understanding and recalling (lower-order thinking) to free brain resources for higher-order thinking (Eppler & Burkhard, 2008; Lee, 2012). See example of graphic organizer of Ana's higher-order thinking in Fig. 1.

Teachers have used graphic organizers to help their students understand the relations among ideas (Larkin & Simon, 1987; Nesbit &

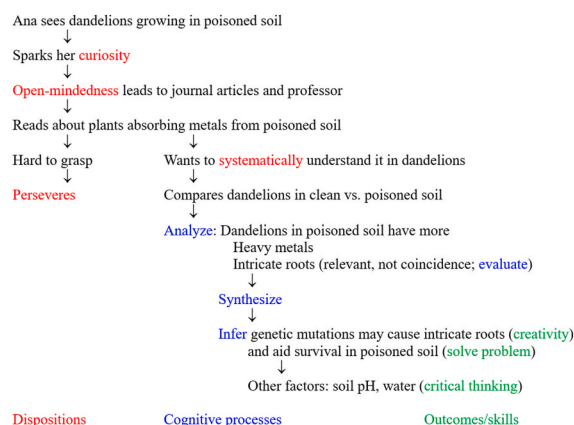


Fig. 1. Graphic organizer of Ana's higher-order thinking dispositions, cognitive processes, and outcomes/skills.

Adesope, 2006) and use their *hippocampi* create corresponding links from these visual stimuli to their *cerebral cortices* (from short-term memory to long-term memory) in their brains. Graphic organizers have boosted students' meaningful learning (Akinsanya & Williams, 2004; Hay, 2007; Romero García et al., 2017) and learning outcomes (Dexter et al. 2011), especially for understanding texts (science: Roman et al., 2016; social studies: Meo, 2008; language learning: Imsa-ard, 2022). Hence, we propose this hypothesis.

H-1. Students using graphic organizers outperform other students on higher-order thinking.

However, some graphic organizers, especially poor ones, can hinder higher-order thinking by worsening cognitive load, split attention, or decrease coherence. Some highlight small details, jumble the order of facts, or mismatch attributes (e.g., color, size, shape) with their meanings (Schnotz & Bannert, 2003). Students might fixate on a complicated graphic organizer (see Fig. 1) instead of listening to the teacher, which can burden their minds with unnecessary details (*excessive cognitive load*, Sweller et al., 2011). Also, the information in the graphic organizer might not align with their learning goals and obstruct their understanding (Schnotz & Bannert, 2003). Even as their teacher lectures, students' thoughts might wander back to the graphic organizer. This can split their attention, break their focus, disrupt the steady flow of information (Hittleman, 1985). Also, some students know the basic ideas but need the details from the lecture. These students can waste time on the graphic organizer, especially one tangled with tangential ideas (less coherence). These can slow learning (*redundancy effect*, Sweller et al., 2011) or cause students to ignore the lecture altogether (Schnotz, 2014).

2.3. Potential moderators

To account for past studies' different results, we test for moderation effects of higher-order thinking measures, graphic organizer types, demographics, and study designs.

2.3.1. Higher-order thinking measures

2.3.1.1. Higher-order thinking components: dispositions, cognitive processes, and skills. Studies assessed higher-order thinking dispositions, cognitive processes, or outcomes/skills. As higher-order thinking dispositions and higher-order thinking cognitive processes precede higher-order thinking outcomes, successful higher-order thinking outcomes require both suitable higher-order thinking dispositions and higher-order thinking cognitive processes (Thamrin & Widodo, 2019; Tikhonova & Kudina, 2015). By contrast, neither the dispositions nor the cognitive processes require higher-order thinking outcomes (Thamrin & Widodo, 2019; Tikhonova & Kudina, 2015). Hence, graphic GO-HOT links are likely larger when assessing higher-order thinking dispositions or cognitive processes rather than higher-order thinking skills.

H-2. The positive link between graphic organizer and higher-order thinking is stronger for dispositions or cognitive processes than skills.

2.3.1.2. Divergent vs. convergent higher-order thinking. Some researchers divide higher-order thinking into (a) logical thinking and assessment, including critical thinking (*convergent*) and (b) original, flexible, creative thinking (*divergent*; Lu et al., 2025; Zhan et al., 2023). As graphic organizers organize ideas with a concise, structured representation of idea (e.g., hierarchies, networks, flow diagrams), they reduce cognitive load, especially LOT and convergent thinking (Ezzeldin, 2017). This frees more resources for more difficult, divergent thinking (Eppler & Burkhard, 2008), creation of new ideas (Leeds et al., 2019), or deeper students' conceptual

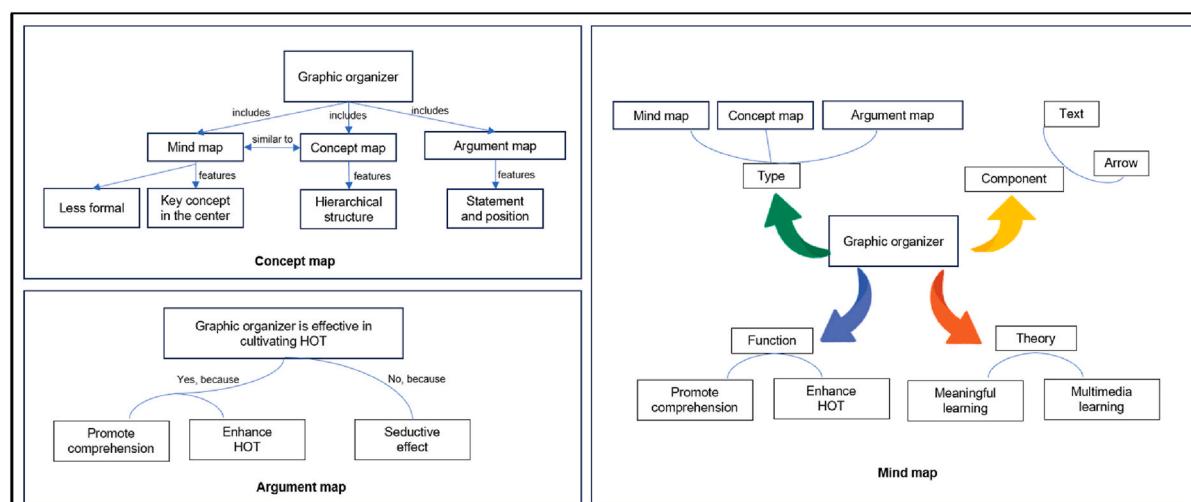


Fig. 2. Illustration of concept map, argument map, and mind map.

understanding (Watson et al., 2016). Hence, graphic organizers might benefit divergent higher-order thinking more than convergent one.

H-3. The positive link between graphic organizer and higher-order thinking is stronger for divergent higher-order thinking than convergent higher-order thinking.

2.3.2. Graphic organizer types

Graphic organizer types include types of maps (e.g., mind, concept, argument; Kansızoglu, 2017), fishbone diagrams, Venn diagrams, flowcharts, and so on (Scott, 2011). However, most studies of graphic organizers and higher-order thinking used mind maps, concept maps or argument maps (see Fig. 2). Mind maps are visual diagrams that organize ideas and information around a central concept, with branches representing related themes and subtopics (Edwards & Cooper, 2010). Often used in brainstorming, students using mind maps identified more relations among ideas and used these relations to remember more, compared to other students (Farrand et al., 2002; Palaniappan et al., 2023). By contrast, concept maps often summarize key ideas and relations, so they are more complex and can become unwieldy compared to the single-concept focus of mind maps (Shi et al., 2023). Indeed, students often learn more from mind maps than concept maps (Abbas et al., 2018; Redhana et al., 2021), though their effects on critical did not differ significantly in Suardana et al. (2020).

Whereas concept maps and mind maps organize ideas and their relations for greater understanding, argument maps structure the logic behind claims, evidence, warrants and backing to persuade others of a specific conclusion (Davies, 2011). Argument maps boost students' critical thinking during structured writing (Botley & Hakim, 2014; Liu et al., 2024) and problem-solving (Hoffmann, 2005). As arguments are typically more structured than brainstorming, we expect the added structure of a mind map to be much more beneficial to brainstorming than an argument map to argumentation. Like mind maps, arguments have clear lines, so both are often more organized than concept maps (Davies, 2011). Indeed, among students studying science, those using argument maps learned more than those using concept maps (Kabataş Memiş & Karakuş, 2021). Hence, we propose the following hypothesis.

H-4. The positive link between graphic organizer and higher-order thinking is strongest for mind maps, weaker for argument maps, and weakest for concept maps.

2.3.3. Demographics

Demographics include national income, age, and year. Countries with higher national income (*gross domestic product [GDP]* per capita) often spend more money to create better technology or train their teachers better (Nelson, 2005). Better technology can yield better graphic organizers. Such graphic organizers can organize key ideas and their relations better to help students learn them more easily. Furthermore, better teachers can use graphic organizers more effectively to help their students learn more. Hence, graphic organizers might have stronger effects on higher-order thinking in richer countries than in poorer ones.

H-5. The positive link between graphic organizer and higher-order thinking is higher in countries with higher GDP per capita.

Age might moderate the effect of graphic organizer on higher-order thinking. As older students' brains are more mature (e.g., formal operations of secondary school students vs. concrete operations of primary school students; Carey et al., 2015), they likely have superior higher-order thinking and representational skills. So, they have more skills that graphic organizers can activate, and can better use graphic organizers. Moreover, unimportant graphic organizer details (e.g., images) more readily distract primary school students and hinder their graphic organizer use, compared to older students (*seductive-details effect*, Rowland-Bryant et al., 2009). Compared to younger students, secondary school students often have stronger (*meta*)-representational skills (diSessa and Sherin, 2000; Pande & Chandrasekharan, 2017; Uttal & Doherty, 2008), so they can use graphic organizers more effectively (Hillmann, 1993). As university students have mature representation skills, they can readily create their own spatial metaphors to pick out key ideas and organize them, reducing the comparative benefit of an external graphic organizer (*self-scaffolding*, Bickhard, 2013). Hence, graphic organizer benefits might be highest among secondary school students and lowest among university students. Indeed, concept maps benefit secondary school students more than university students (Nesbit & Adesope, 2006).

H-6. The positive link between graphic organizer and higher-order thinking is strongest for secondary school students, tapering off for primary school students, and lowest among university students.

Later years often bring more advanced technology (Fraillon et al., 2014) and progress toward better teaching methods (Compayre & Payne, 2015). Advanced technology can yield more effective graphic organizers. Also, teachers using better teaching methods can help their students use a graphic organizer more effectively. Hence, we expect studies published in later years to show larger effect sizes of graphic organizers on higher-order thinking.

H-7. The positive link between graphic organizer and higher-order thinking is larger in later publication years.

2.3.4. Other moderators

We do not expect gender, academic subject, or study duration to moderate the link between graphic organizer and higher-order thinking. Indeed, gender did not moderate the link between concept mapping and critical thinking in a prior analysis (Barta et al., 2022). Also, meta-analyses of mind mapping and cognitive outcomes showed no significant academic subject moderations (Nesbit & Adesope, 2006; Shi et al., 2023) or study duration moderations (Shi et al., 2023). Still, following other meta-analyses, we test for their possible moderation effects.

2.4. This study

Past studies of graphic organizers and higher-order thinking showed mixed results. So, this meta-analysis adjudicates the overall effect of the independent variable graphic organizer on the dependent variable higher-order thinking and tests for moderators (graphic organizer, higher-order thinking, demographics, etc.) that explain the different results.

3. Methods

3.1. Literature search and coding

3.1.1. Literature search

We identified relevant primary studies by searching seven databases: Google Scholar, Web of Science, Elsevier SDOL, EBSCO, Springer, PsycINFO and ProQuest Digital Dissertations. We used the following combination of search terms: (“diagram” OR “graphic organizer” OR “mind mapping” OR “concept mapping” OR “argument mapping” OR “cognitive mapping”) AND (“critical thinking” OR “higher-order thinking” OR “reasoning” OR “creative thinking” OR “problem solving” OR “argumentation” OR “argumentative thinking” OR “decision-making” OR “metacognitive thinking” OR “creativity”). If an original study was not publicly available, we tried to get it from a school library or by emailing the authors. We completed this search process in January 2025, retrieving 665 articles.

3.1.2. Inclusion criteria

We applied the following inclusion criteria to these studies: (a) empirically tested the relation between the independent variable graphic organizer use and the dependent variable higher-order thinking; (b) had a quasi-experimental, experimental, or pre-post design; (c) had sufficient statistical information to calculate the effect size (e.g., sample sizes, means, standard deviations) (See Fig. 3). After applying these criteria, 66 articles remained.

3.1.3. Quality assessment

Two authors independently assessed the quality of each study with Kmet, Lee, and Cook's (2004) fourteen-item checklist. These items helped assess the internal validity of quantitative research across multiple methodological dimensions. One such item is the clarity and completeness of the research question/objective. Each item was rated as “yes” (2 points), “partial” (1 point), or “no” (0 points) for a maximum of 28 points. To accommodate the heterogeneity of quantitative study designs, some items can be designated “not applicable” (N/A). All 66 original studies had at least moderate quality (over 50%; $M = 23.75$ [85%]) and were included in this

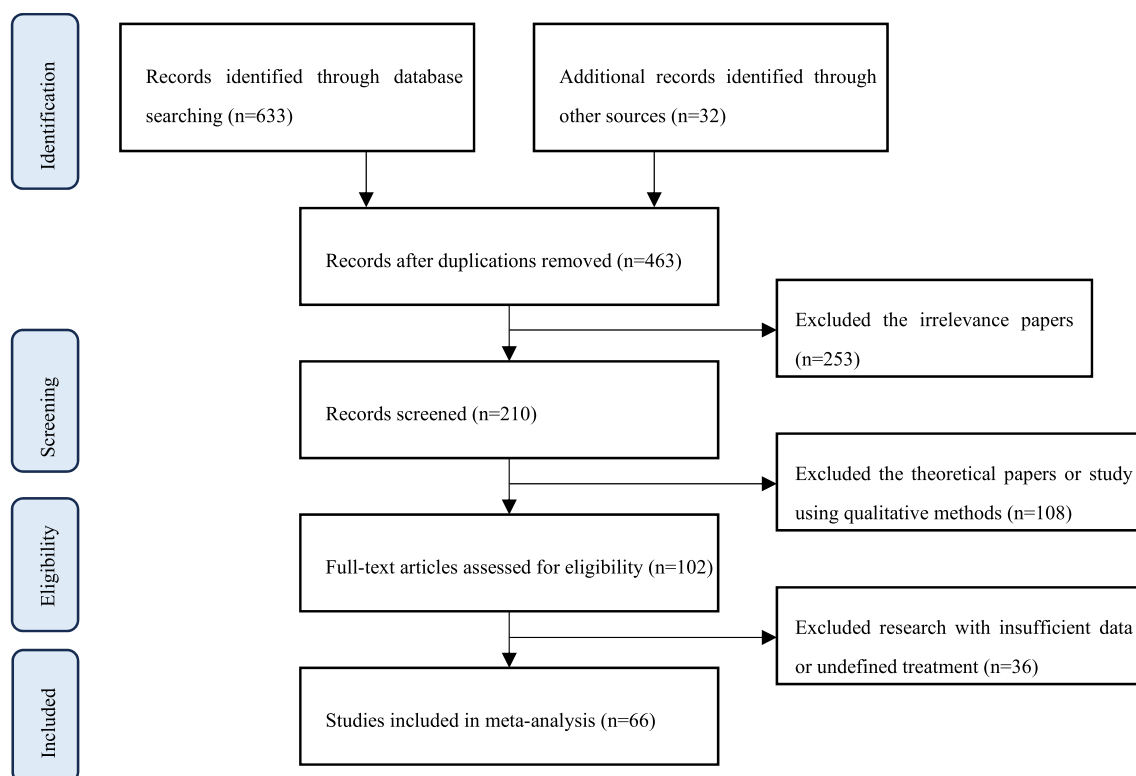


Fig. 3. Flow diagram of literature search and study inclusion criteria.

meta-analysis.

3.1.4. Study feature coding

For each study, two authors coded its features (Lipsey & Wilson, 2001): study name, publication year, sample size, effect size, GDP per capita, grade level, gender, higher-order thinking measures (disposition, cognitive process vs. skill; convergent vs. divergent), graphic organizer type (mind map, concept map, argument map), academic subject, and study duration (see Table 1). In one study (Su, 2018), the teacher created the graphic organizer and gave it to the students. In all other studies, the students created the graphic organizer. A common sequence in graphic-organizer-based learning activities across the studies involved teachers introducing the basic knowledge of the graphic organizer used in the intervention, students generating their own graphics, and teachers providing feedback. Krippendorff's (2011) α between the two authors' ratings is 0.813, showing acceptable inter-rater reliability.

Researchers measured higher-order thinking dispositions, cognitive processes, and skills in different ways. To measure higher-order thinking disposition, researchers in all 34 primary studies used Likert-type scales (strongly disagree to strongly agree) with questions such as "I like to ask something that no one has imagined before" (e.g. Chiu & Hwang, 2024). Meanwhile, researchers in all 17 studies measuring cognitive processes asked participants to solve open problems via writing. Then, they used scoring rubrics to capture these cognitive processes (Lane, 1993). For example, Khrais & Saleh (2020) asked students to read a letter with flawed reasoning and then write an essay assessing the claims in the letter. Then, the researchers applied a rubric to assess participants' quality of reasoning (especially counter-arguments), relevance and clarity of critiques, and use of evidence. Lastly, researchers in all 57 studies used multiple-choice tests to measure higher-order thinking skills. These include standardized tests, like the California Critical Thinking Skills Test (CCTST, Facione, 1990). An example item is "the happiness level in some countries can be higher or lower than in others with the same personal income level. The statement is: (a) True; (b) Probably True; (c) Insufficient Data; (d) Probably False; (e) False." Studies in the field of nursing in US used the computerized exams by Health Education Systems, Inc. (Morrison et al., 2004). Other researchers developed and validated multiple-choice questions with high reliability to test students' higher-order thinking skills. (e.g., Min et al., 2023). We test whether these different measures of the outcome variable higher-order thinking (disposition, cognitive process, skill) moderate the relation between graphic organizer and higher-order thinking (Chiu & Roberts, 2018).

3.2. Data processing and analysis

3.2.1. Effect size computation

For each study, we computed the effect size Hedge's g (Vøllestad et al., 2012). If available, we used the sample sizes, means, and standard deviations in either (1) the post-test data of the experimental and control groups or (2) the pre-test and post-test data of the single experiment. Otherwise, we used χ^2 , t , or F values.

3.2.2. Homogeneity test

We computed Cochran's Q and I^2 to test for heterogeneity in the effect sizes across studies (Huedo-Medina et al., 2006). Substantial heterogeneity (significant Q ; $I^2 > 75$) requires *random-effects models* and moderation tests (Lipsey & Wilson, 2001).

3.2.3. Publication bias

To evaluate publication bias, we made a funnel plot (Lin & Chu, 2018) and ran Egger et al.'s (1997) regression. A sharply asymmetric funnel plot or a significant, non-zero regression intercept indicates publication bias.

3.2.4. Data analysis

We used the *meta*, *dmetar* and *metafor* packages in R (version 2022.02.0 + 443) for our meta-analysis (Schwarzer et al., 2015). A three-level *meta-analytic variance components* model (Assink & Wibbelink, 2016) showed no significant variance between effect sizes within studies (level 2), so we used the traditional two-level meta-analysis. Two random-effects models calculated the effect sizes and the homogeneity test, followed by moderator tests. Wald tests determined whether differences in effect sizes were significant (Hansen, 2022).

4. Results

This meta-analysis examined 72 effect sizes from 66 articles of 4824 participants in 19 countries from 2000 to 2024. Each study's sample size ranged from 20 to 410 (mean = 67).

4.1. Homogeneity test and effect size

The significant Q statistic ($p < .001$) and large I^2 (84%) indicate substantial heterogeneity. So, we used a random-effects model. It showed a positive link between graphic organizer and higher-order thinking, supporting hypothesis H-1 ($g = 0.778$, $k = 72$, 95% CI = [0.642, 0.915]) (see Table 2).

4.2. Assessment of publication bias

The distribution of the 72 effect sizes in the funnel plot was mostly symmetric (see Fig. 4), and Egger's regression (see Table 3) did

Table 1

Studies included in the meta-analysis.

Author (Year)	N	Hedge's g	SE	GDP per capita Z	Academic Section	HOT Component	HOT Type	Subject	Intervention Duration	GO type	Gender	Quality Assessment
Achor et al. (2024)	175	2.158	0.200	−0.923	S	C	C	STEM	NA	other	0.51	24
Akinoglu (2013)	60	0.834	0.266	−0.470	P	S	C	STEM	4-12	Con	NA	22
Alfayoumi (2019)	40	0.448	0.224	−0.805	U	D	C	STEM	NA	Con	NA	23
Atay & Karabacak (2012)	80	1.237	0.242	−0.470	U	D	C	STEM	≤4	Con	NA	22
Ayal et al. (2016)	130	1.215	0.190	−0.796	S	S	C	STEM	>12	Mnd	NA	19
Bilik et al. (2020)	410	0.172	0.099	−0.470	U	D	C	STEM	NA	Con	0.23	23
Bin et al. (2022)	38	1.837	0.272	−0.537	U	D	D	NA	≤4	Mnd	NA	23
Bixler et al. (2015)	27	0.250	0.269	1.701	U	S	C	STEM	≤4	Con	NA	22
Boyadjian-Samawi (2006)-1	77	0.153	0.229	1.701	U	S	C	STEM	>12	Con	0.12	24
Boyadjian-Samawi (2006)-2	77	0.286	0.230	1.701	U	D	C	STEM	>12	Con	0.12	24
Carvalho et al. (2020)-1	21	0.148	0.424	−0.328	U	S	C	STEM	NA	Con	NA	20
Carvalho et al. (2020)-2	56	0.225	0.265	−0.328	U	S	C	STEM	NA	Con	NA	20
Carwie (2009)	98	−0.020	0.203	1.701	U	S	C	STEM	4-12	Arg	0.19	24
Chen & Hwang (2020)	37	0.346	0.232	−0.537	U	D	C	HSS	4-12	Con	NA	26
Chen (2020)-1	30	0.900	0.268	−0.537	S	D	D	HSS	>12	Mnd	0.17	26
Chen (2020)-2	30	1.848	0.306	−0.537	S	C	D	HSS	>12	Mnd	0.17	26
Chen et al. (2011)	95	0.350	0.205	−0.537	U	S	C	STEM	>12	Con	NA	26
Chiu & Hwang (2024)-1	48	1.095	0.305	−0.537	U	D	C	NA	4-12	Mnd	NA	28
Chiu & Hwang (2024)-2	48	1.026	0.303	−0.537	U	D	D	NA	4-12	Mnd	NA	28
Crudele & Raffaghelli (2023)	103	1.066	0.148	0.294	U	C	C	NA	≤4	Arg	NA	28
Cullen et al. (2018)	105	0.738	0.142	1.701	U	C	C	NA	4-12	Arg	NA	18
D'Antoni et al. (2010)	131	0.194	0.174	1.701	U	S	C	STEM	≤4	Mnd	0.47	24
Dwyer et al. (2011)	53	0.586	0.279	2.365	U	S	C	NA	4-12	Arg	NA	27
Dwyer et al. (2012)	42	0.883	0.227	2.365	U	C	C	NA	4-12	Arg	NA	28
Eggert et al. (2017)	20	0.170	0.311	0.775	S	C	C	STEM	≤4	Con	NA	27
Elasrag & Elsabagh (2020)	100	1.145	0.152	0.135	U	D	C	STEM	NA	Mnd	0.2	24
Esfandiari et al. (2021)	30	0.569	0.260	−0.800	NA	S	C	HSS	NA	Arg	1	26
Esquillo (2023)	56	2.043	0.327	−0.825	S	D	C	STEM	NA	Arg	NA	25
Hazaymeh & Alomery (2022)	42	0.986	0.321	0.577	U	S	C	HSS	NA	Mnd	NA	20
Heglund (2015)	18	0.246	0.327	1.701	S	S	C	STEM	4-12	Arg	0.94	27
Hu & Hwang (2024)-1	56	0.478	0.268	−0.537	U	D	C	NA	≤4	Con	NA	24
Hu & Hwang (2024)-2	56	1.151	0.285	−0.537	U	D	D	NA	≤4	Con	NA	24
Huang et al. (2017)	57	1.056	0.199	−0.537	P	S	C	NA	>12	Con	0.46	26
Huang et al. (2012)-1	134	0.932	0.181	−0.537	U	D	C	STEM	>12	Con	NA	27
Huang et al. (2012)-2	134	0.507	0.175	−0.537	U	S	C	STEM	>12	Con	NA	27
Hwang et al. (2021)	40	0.152	0.311	−0.537	S	D	C	STEM	4-12	Con	NA	25
Israel et al. (2020)	75	0.224	0.156	1.701	U	S	C	STEM	4-12	Mnd	0.18	21
Kaddoura et al. (2016)	83	0.824	0.227	1.701	U	S	C	STEM	NA	Con	0.2	21
Kamble & Tembe (2013)	42	0.583	0.221	−0.864	U	C	D	STEM	4-12	Con	NA	22
Kharismayuni et al. (2021)	50	0.666	0.286	−0.796	S	S	NA	STEM	NA	other	NA	21
Khodadady & Ghanizadeh (2011)	36	1.150	0.353	−0.800	U	S	C	HSS	4-12	Con	0.14	26
Khrais & Saleh (2020)	58	0.748	0.191	−0.805	U	C	C	STEM	4-12	Con	0.3	19
Laelasari & Anggraeni (2017)	74	1.721	0.272	−0.796	S	C	D	STEM	≤4	other	NA	17
Lazuardini et al. (2019)	33	1.509	0.276	−0.796	S	S	D	STEM	NA	Mnd	NA	24
Lee et al. (2013)	94	0.318	0.206	−0.537	U	S	C	STEM	>12	Con	0	24
Maldonado (2014)	40	0.180	0.222	1.701	U	S	C	STEM	>12	Con	NA	25
Maryam et al. (2021)	100	0.594	0.205	−0.800	U	S	C	STEM	>12	Con	0.52	25

(continued on next page)

Table 1 (continued)

Author (Year)	N	Hedge's g	SE	GDP per capita_Z	Academic Section	HOT Component	HOT Type	Subject	Intervention Duration	GO type	Gender	Quality Assessment
Min et al. (2023)	60	0.794	0.268	−0.537	S	S	C	HSS	>12	other	NA	25
Mohamed El-Sayed et al. (2023)	156	2.348	0.208	−0.845	U	D	C	STEM	NA	Mnd	0	25
Mutiara et al. (2021)	59	0.430	0.260	−0.796	S	S	C	STEM	NA	Mnd	NA	17
Ngajie et al. (2020)	39	0.443	0.227	−0.537	U	C	C	HSS	4-12	Arg	0.05	26
Nirmala & Shakuntala (2011)	84	0.172	0.217	−0.864	U	S	C	STEM	4-12	Con	NA	18
Pinandito et al. (2023)	54	1.177	0.208	−0.796	U	S	NA	STEM	≤4	Con	NA	22
Ristanto et al. (2022)	33	1.064	0.260	−0.796	S	C	C	STEM	NA	Arg	NA	25
Silva et al. (2022)	33	0.697	0.354	−0.057	U	C	NA	NA	>12	Con	0.15	25
Simper et al. (2016)	28	0.112	0.367	0.728	U	C	D	HSS	≤4	Con	0.11	20
Su & Long (2021)	72	−0.613	0.239	−0.537	U	C	C	NA	≤4	Con	0.29	26
Su (2018)	28	0.887	0.277	−0.537	U	NA	C	STEM	NA	Con	NA	24
Sun et al. (2019)	49	2.132	0.252	−0.537	S	C	D	STEM	NA	other	0.57	24
Tian et al. (2023)	49	0.991	0.213	−0.537	U	D	C	HSS	≤4	Con	NA	19
Turkestani et al. (2024)	79	0.464	0.226	0.135	U	S	C	STEM	>12	Mnd	NA	22
Twyman (2003)	50	0.781	0.289	1.701	S	C	D	HSS	4-12	Con	0.5	25
Uçar & Demiraslan Çevik (2020)	43	0.825	0.313	−0.470	U	C	C	STEM	4-12	Arg	NA	28
van Boxtel & Veerman (2001)	38	0.908	0.335	1.158	S	C	C	STEM	≤4	Con	NA	23
Wang & Liao (2014)	41	1.373	0.342	−0.537	U	D	C	HSS	4-12	Con	0.46	26
Wei et al. (2019)	28	1.222	0.288	1.701	P	C	C	HSS	>12	other	0.46	26
Wheeler & Collins (2003)	76	0.112	0.230	1.701	U	S	C	STEM	4-12	Con	0.05	22
Widiana & Jampel (2016)	20	1.271	0.341	−0.796	P	C	D	STEM	NA	Mnd	NA	24
Wu & Wu (2020)	64	0.382	0.294	−0.537	U	D	C	STEM	NA	Mnd	0.03	27
Wu et al. (2016)	52	1.121	0.177	−0.537	U	C	C	STEM	≤4	other	NA	24
Yang & Yin (2015)	90	0.158	0.209	−0.537	P	C	D	STEM	4-12	other	NA	18
Yilmaz-Na & Sönmez (2023)	60	1.289	0.281	−0.470	U	D	D	HSS	>12	Arg	NA	27

GO type: argument mapping = Arg, cognitive mapping = Cog, mind mapping = Mnd, other.

Measure of HOT: disposition = D, skills = S, cognitive process = C.

HOT type: convergent = C, divergent = D.

Subject: humanity and social studies = HSS, STEM.

Grade level: primary = P, secondary = S, university = U.

Study duration in weeks: ≤4, 4-12, >12.

Gender: male/all.

Table 2

Random-model of the effect of graphic organizer on students' higher order thinking.

	k	g	95%-CI	t	p-value	Test of heterogeneity		
						Q	p-value	I ² (%)
Graphic organizer	72	0.778	[0.642, 0.915]	11.36	<0.001	443.56	<0.001	84.0

not have a significant intercept ($t = 1.841$, $p = .070$). Hence, we have no evidence of publication bias.

4.3. Moderator analysis

Higher-order thinking measures, graphic organizer measures, and demographics moderated the positive link between graphic organizer and higher-order thinking (see [Tables 4 and 5](#)).

4.3.1. Higher-order thinking and graphic organizer measures

Higher-order thinking components moderated the GO-HOT link ($Q_{BE} = 8.66$, $p < .05$). This link was strongest for dispositions ($g = 0.945$, 95% CI = [0.670, 1.221]) and for cognitive process ($g = 0.912$, 95% CI = [0.631, 1.194]), and much smaller for skills ($g = 0.560$, 95% CI = [0.411, 0.708]). Wald tests showed significant differences between the effect size for skills and for both cognitive processes ($W = 137.48$; $p < .001$) and dispositions ($W = 53.47$; $p < .001$), supporting [H-2](#). The effects for dispositions and skills did not differ significantly ($W = 0.39$; $p = .533$).

Higher-order thinking type also moderated the GO-HOT link ($Q_{BE} = 7.07$, $p < .01$). The effect is stronger in divergent thinking ($g = 1.167$, 95%CI = [0.841, 1.493]) than convergent thinking ($g = 0.680$, 95%CI = [0.535, 0.825]). Hence, this result support [H-3](#).

Graphic organizer type moderated the GO-HOT link ($Q_{BE} = 9.13$, $p < .05$). This link was largest for mind mapping ($g = 1.041$, 95% CI = [0.704, 1.379]), tapering for argument mapping ($g = 0.798$, 95% CI = [0.469, 1.237]), and concept mapping ($g = 0.548$, 95% CI = [0.704, 1.379]). All effect size differences were significant ($W_{\text{mind, argument}} = 12.52$, $p < .001$; $W_{\text{argument, concept}} = 19.25$, $p < .001$; $W_{\text{mind, concept}} = 70.21$, $p < .001$), supporting [H-4](#).

4.3.2. Demographics

GDP per capita, grade level, and publication year showed significant moderation. The positive link was larger in countries with higher GDP per capita ($Q_{(1, 70)} = 9.43$, $p < .01$), supporting [H-5](#).

Grade level also moderated the GO-HOT link ($Q_{BE} = 6.61$, $p < .05$). The link was strongest among secondary students ($g = 1.113$, 95% CI = [0.759, 1.467]), weaker for primary school students ($g = 0.877$, 95% CI = [0.303, 1.451]) and weakest for university

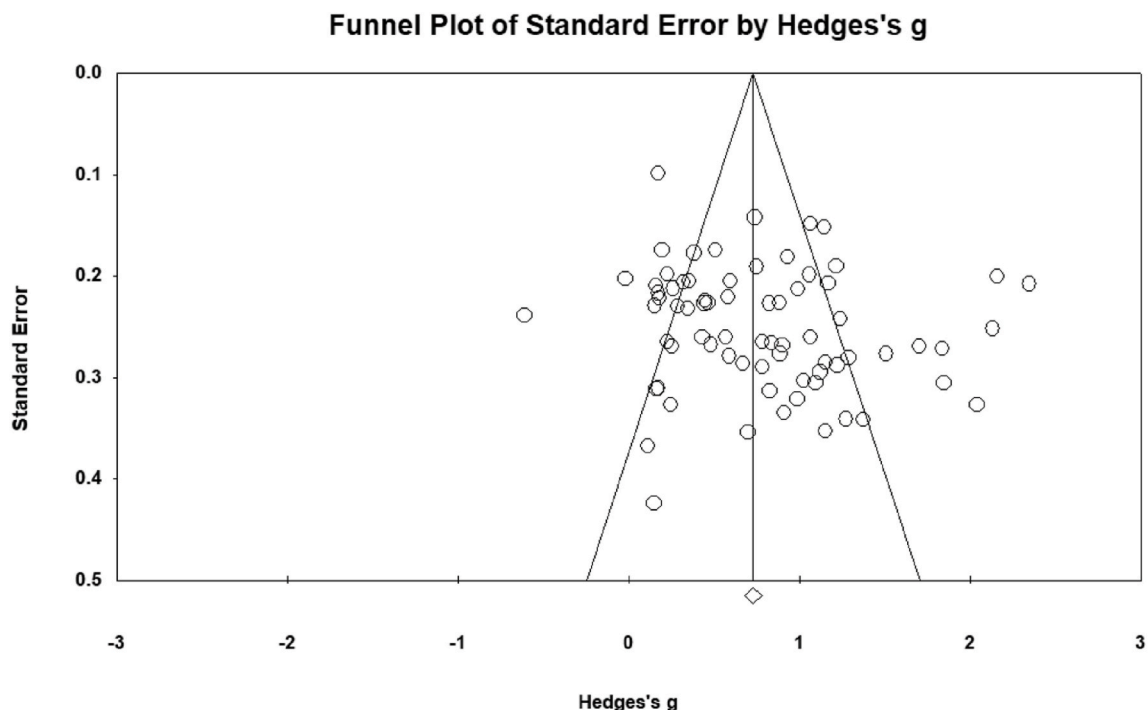


Fig. 4. Funnel plot of the studies included in the meta-analysis.

Table 3

Egger's test for studies included in the meta-analysis.

	Intercept	SE	95% CI	t	p
Graphic organizer	1.965	1.067	[-0.164 4.094]	1.841	0.070

Table 4

Moderators of Graphic Organizers and Higher Order Thinking: Univariate analyses (categorical variables).

Group	Moderator	Between group effect (Qbet)	k	g	SE	95% CI	Heterogeneity test within groups (Q _w)	I ² (%)
Higher Order	HOT component	8.66*					443.23***	
	Disposition		20	0.945	0.140	[0.670, 1.221]		88.7
Thinking (HOT)	Skill		29	0.560	0.076	[0.411, 0.708]		67.8
	Process		22	0.912	0.144	[0.631, 1.194]		86.7
	HOT type	7.07**					438.78***	
	Convergent		55	0.680	0.074	[0.535, 0.825]		83.6
	Divergent		14	1.167	0.166	[0.841, 1.493]		82.2
Graphic Organizer (GO)	GO type	9.13*					332.03***	
	Argument mapping		12	0.798	0.150	[0.469, 1.237]		75.6
	Concept mapping		36	0.548	0.071	[0.403, 0.692]		69.6
	Mind mapping		16	1.041	0.158	[0.704, 1.379]		88.0
Demographics	Grade level	6.61*					443.19***	
	Primary		5	0.877	0.207	[0.303, 1.451]		73.5
	Secondary		17	1.113	0.167	[0.759, 1.467]		84.4
	University		49	0.659	0.074	[0.509, 0.808]		81.7
Other	Subject	1.35 (p = .246)					380.57***	
	Humanities		14	0.903	0.120	[0.645, 1.161]		58.9
	STEM		47	0.728	0.091	[0.545, 0.912]		86.9
	Study duration	2.33 (p = .313)					215.35***	
	Less than 4 weeks		15	0.791	0.172	[0.421, 1.160]		56.6
	4 weeks to 12 weeks		21	0.563	0.084	[0.389, 0.738]		72.4
	More than 12 weeks		17	0.735	0.111	[0.500, 0.971]		86.0

*p < .05; **p < .01; ***p < .001.

students ($g = 0.659$, 95% CI = [0.509, 0.808]). All effect size differences were significant ($W_{\text{secondary, primary}} = 6.10$, $p = .014$; $W_{\text{primary, university}} = 60.04$, $p < .001$; $W_{\text{secondary, university}} = 63.77$, $p < .001$), supporting H-6.

The positive GO-HOT link was larger in later publication years, supporting H-7 ($Q = 7.01$, $p < .05$). Gender, academic subject, and study duration did not moderate the GO-HOT link ($Q_{\text{gender}} = 0.36$, $p = .553$; $Q_{\text{subject}} = 1.35$, $p = .246$; $Q_{\text{duration}} = 2.33$, $p = .313$).

5. Discussion

Unlike single studies with small samples in single contexts, this meta-analysis of 72 effect sizes from 66 studies of 4824 students in 19 countries showed a positive link between graphic organizer and higher-order thinking. This result extends those of past meta-analyses showing that graphic organizer is linked to greater learning outcomes (Dexter et al., 2011), especially learning language (Kansizoglu, 2017). Unlike mono-method, single-site studies that did not test for moderators across interventions or contexts, this meta-analysis showed that higher-order thinking component, higher-order thinking type, graphic organizer type, GDP per capita,

Table 5

Moderators of Graphic Organizers and Higher Order Thinking: Univariate analyses for continuous variables (random-effects model).

	Parameter	Estimate	SE	95%CI for B
Gender	β_0	0.643	0.208	[0.216, 1.069]
	β_1	0.325	0.540	[-0.784, 1.434]
	F (df1 = 1, df2 = 27) = 0.362 (p = .553)			
GDP per capita	β_0	0.778	0.065	[0.649, 0.907]
	β_1	0.197	0.064	[0.325, 0.069]
	F (df1 = 1, df2 = 70) = 9.430**			
Publish year	β_0	-62.977	24.086	[-111.016, -14.938]
	β_1	0.032	0.012	[0.008, 0.055]
	F (df1 = 1, df2 = 70) = 7.006*			

*p < .05; **p < .01; ***p < .001.

grade level and publication year moderated the link between graphic organizer and higher-order thinking.

5.1. Overall positive GO-HOT link

The positive link between graphic organizer and higher-order thinking supports hypothesis H-1. This result fits the view that graphic organizers use spatial metaphors (Lakoff & Johnson, 2024) to select key ideas/relations and organize them, which reduces the cognitive load (Mayer, 2002) of lower-order thinking and lets more brain processes engage in higher-order thinking (Eppler & Burkhard, 2008; Lee, 2012). This result also aligns with past studies showing how teachers used graphic organizers to help their students learn new concepts (Akinsanya & Williams, 2004; Hay, 2007; Romero García et al., 2017) and understand accompanying texts (Imsa-ard, 2022; Meo, 2008; Roman et al., 2016).

5.2. Higher-order thinking

The GO-HOT link was much stronger for higher-order thinking dispositions and cognitive processes than for higher-order thinking outcomes/skills, supporting H-2. This result coheres with the idea that higher-order thinking outcomes require both dispositions and cognitive processes, whereas neither of them require higher-order thinking outcomes (Thamrin & Widodo, 2019; Tikhonova & Kudinova, 2015). Hence, this moderation result highlights the importance of distinguishing higher-order thinking outcomes from dispositions and cognitive processes in theoretical models, empirical studies, and meta-analyses. Whether higher-order thinking dispositions and cognitive processes largely co-occur or not requires future studies that capture both phenomena.

Furthermore, GO-HOT link was larger for divergent higher-order thinking than convergent higher-order thinking, supporting Hypothesis 3. This result fits the view that graphic organizers help organize ideas to reduce cognitive load, especially the need for lower order thinking skills and effort for convergent thinking (Ezzeldin, 2017). This frees up brain resources for divergent thinking (Eppler & Burkhard, 2008), such as creation of new ideas (Leeds et al., 2019).

5.3. Graphic organizer

Unlike past studies that examined only one type of graphic organizer (e.g., Shi et al., 2023), this meta-analysis showed the relative effects of different graphic organizers. Specifically, the GO-HOT link was largest for mind maps, much smaller for argument maps, and smallest for concept maps, supporting H-4. These results offer evidence to fit the view that the simpler mind-map organization substantially boosts relations among ideas during brainstorming (Davies, 2011), compared to the more demanding, complex argument maps and concept maps (Shi et al., 2023). These results also align with those of past studies showing that students often learn more using mind maps rather than concept maps (Abbas et al., 2018; Redhana et al., 2021).

Furthermore, the larger GO-HOT link for argument maps than concept maps coheres with the view that the former is simpler and easier to use (Davies, 2011). Argument maps typically have a linear, logical or temporal structure, so they have simpler organizations than concept maps, which can have any structure and thus be unwieldy (Davies, 2011). This result is also consistent with those of past studies showing that students learn more science using argument maps than concept maps (Kabataş Memiş & Karakuş, 2021).

5.4. Demographics

Extending past studies, this meta-analysis showed that country, publication year, and age moderated the GO-HOT link. Studies in richer countries or later years showed larger GO-HOT links, supporting H-5 and H-7. These results fit the views that richer countries spend more money to create better graphic organizers or train better teachers to use them more effectively to help their students learn more (Nelson, 2005). Likewise, later years often bring (a) more advanced technology (Fraillon et al., 2014) for better graphic organizers and (b) better trained teachers (Compayre & Payne, 2015) who can better use them to improve their students' learning.

Extending past studies, this meta-analysis showed that the GO-HOT link was strongest among secondary school students, much weaker for primary school students, and weakest among university students, supporting H-6. This result not only aligns with that of a meta-analysis showing that concept maps benefit secondary students more than university students (Nesbit & Adesope, 2006) but also extends it by showing the relative effect size of primary school students. The weakest GO-HOT link among university students fits the view that their strong representation skills let them create their own graphic organizers, so they do not need an external graphic organizer for their higher-order thinking as much as younger students do. The results for the younger students cohere with the view that compared to primary school students, secondary school students use graphic organizers more effectively. Specifically, they are less distracted by unimportant graphic organizer details (*seductive-details effect*, Rowland-Bryant et al., 2009) and their stronger (*meta-representational skills*) lets them use graphic organizers more effectively than primary school students (diSessa and Sherin, 2000; Hillmann, 1993; Pande & Chandrasekharan, 2017).

Gender, academic subject, and study duration did not moderate the GO-HOT link. These non-significant results are consistent with those of earlier meta-analyses (Barta et al., 2022; Nesbit & Adesope, 2006; Shi et al., 2023).

5.5. Implications

These results have implications for theory, methodology and practice. For theory, these results show that any comprehensive theory of higher-order thinking must include available tools (e.g., graphic organizer), dispositions, cognitive processes, and outcomes/skills within an ecological model with individual ages, country contexts, and time. Similarly, a rigorous theory of graphic organizers must distinguish among its different types (e.g., mind map vs. argument map vs. concept map), their affordances, disadvantages, and conditions of use.

Methodologically, researchers must separately measure higher-order thinking dispositions, cognitive processes and skills. Also, they can test the degree to which higher-order thinking dispositions co-occur with cognitive processes. As these meta-analysis results showed differences across student ages, regions, and years, researchers should consider how the local ecology might affect their study and thereby design it accordingly.

Practically, this meta-analysis suggests that students can use graphic organizers to boost their higher-order thinking. Hence, teachers can use these results to inform their use of graphic organizers. As graphic organizers raise higher-order thinking dispositions and cognitive processes much more than higher-order thinking outcomes/skills, teachers can scaffold their students to help them improve their higher-order thinking skills. As graphic organizers were more effective for secondary school students and in richer countries, educators can support graphic organizer use for them first, before bringing them to other students. As the GO-HOT link was stronger in later years, students might benefit even more from using graphic organizers in the future.

6. Limitations and implications

This meta-analysis has three primary limitations: studies of few types of graphic organizers, sparse details in the original studies, measures, and language of publication. As we found studies of only three types of graphic organizers and higher-order thinking, these results might not apply to other types of graphic organizers (e.g., fishbone diagrams). Hence, future primary studies can test whether other types of graphic organizers affect higher-order thinking, which will flow into more comprehensive meta-analyses later.

Second, sparse details in the primary studies preclude many moderation tests. For example, many studies lacked details about the students (e.g., prior achievement) and their teachers (e.g., years of experience), so future studies should report this information, which can enrich future meta-analyses.

Third, these primary studies used (a) participants' Likert-type scales to measure their higher-order thinking dispositions, (b) their writing to measure their higher-order thinking cognitive processes, and (c) their multiple-choice test responses to measure their higher-order thinking skills. So, these different methodologies might affect the study results. Hence, future studies can apply the same methodology or multiple methods to measure all outcomes. Also, primary studies mostly used three graphic organizers: mind maps, concept maps, or argument maps. As future primary studies examine flowcharts, Venn diagrams, fishbone diagrams and other graphic organizers, future meta-analyses can test whether these effects hold for different graphic organizers.

Lastly, this meta-analysis only included English publications. Authors of future meta-analysis who can read other languages can include primary studies not published in English.

7. Conclusion

This meta-analysis of 72 effect sizes from 66 studies of 4824 students from 19 countries showed that graphic organizers were positively linked to higher-order thinking ($g = 0.778$). This link was stronger for higher-order thinking dispositions and conceptual processes than for higher-order thinking skills; for mind maps, tapering for argument maps, and concept maps; for divergent higher-order thinking than convergent higher-order thinking; in counties with higher GDP per capita, for secondary school students, tapering for primary school students and university students; and in later publication years.

Declaration of authors' individual contributions

Hao Lei made substantial contributions to the conception or design of the work; AND acquired, analyzed and interpreted data for the work; AND drafted the work and revised it critically for important intellectual content.

Yuxin Ding made substantial contributions to the conception or design of the work; AND acquired, analyzed and interpreted data for the work; AND drafted the work and revised it critically for important intellectual content.

Ming Chiu revised it critically for important intellectual content; AND provided polishing services for English manuscript; AND approved the final version to be published.

All authors read and approved the final manuscript.

Author statement

Hao Lei: I have made substantial contributions to the conception and design of the work; AND acquired, analyzed and interpreted data for the work; AND revised the manuscript critically for important intellectual content.

I agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Yuxin Ding: I have made substantial contributions to the conception or design of the work; AND acquired, analyzed and interpreted data for the work; AND drafted the work and revised it critically for important intellectual content.

I agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Ming Chiu: I have revised it critically for important intellectual content; AND provided polishing services for English manuscript; AND approved the final version to be published.

All persons who have made substantial contributions to the work reported in the manuscript, including those who provided editing and writing assistance but who are not authors, are named in the Acknowledgments section of the manuscript and have given their written permission to be named. If the manuscript does not include Acknowledgments, it is because the authors have not received substantial contributions from non-authors.

Funding

Key Project of the National Social Science Fund for Education of China (Grant Number: APA240021).

Declaration of competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Acknowledgements

All authors read and approved the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.edurev.2026.100822>.

Data availability

Data will be made available on request.

References

- Abbas, S. S., Eldin, A. S., & Elsayed, A. (2018). The effect of concept mapping and mind mapping utilization on students' understanding level: An empirical study. In *Proceedings of the eight international conference on concept mapping*. Medellin, Colombia.
- * Achor, E. E., Jack, G. U., & Uzomah, T. N. (2024). Enhancing students' critical thinking in physics using graphic organizer enhanced and context-based learning strategies. *Journal of Research in Science and Mathematics Education*, 3(1), 1–20.
- * Akinoglu, O. (2013). Effects of concept maps on students' critical thinking skills in science education. *Journal of Environmental Protection and Ecology*, 14(3A), 1424–1431.
- Akinsanya, C., & Williams, M. (2004). Concept mapping for meaningful learning. *Nurse Education Today*, 24(1), 41–46.
- * Alfayoumi, I. (2019). The impact of combining concept-based learning and concept-mapping pedagogies on nursing students' clinical reasoning abilities. *Nurse Education Today*, 72, 40–46.
- Arviani, F. P., Wahyudin, D., & Dewi, L. (2023). Role of teaching strategies in promoting students' higher order thinking skills and critical thinking dispositions. *International Journal of Learning, Teaching and Educational Research*, 22(9), 19. <https://doi.org/10.26803/ijlter.22.9.19>
- Assink, M., & Wibbelink, C. J. (2016). Fitting three-level meta-analytic models in R: A step-by-step tutorial. *The Quantitative Methods for Psychology*, 12(3), 154–174.
- * Atay, S., & Karabacak, Ü. (2012). Care plans using concept maps and their effects on the critical thinking dispositions of nursing students. *International Journal of Nursing Practice*, 18(3), 233–239.
- * Ayal, C. S., Kusuma, Y. S., Sabandar, J., & Dahlan, J. A. (2016). The enhancement of mathematical reasoning ability of junior high school students by applying mind mapping strategy. *Journal of Education and Practice*, 7(25), 50–58.

- Barta, A., Fodor, L. A., Tamas, B., & Szamoskozi, I. (2022). The development of students critical thinking abilities and dispositions through the concept mapping learning method-A meta-analysis. *Educational Research Review*, 37, Article 100481.
- Bickhard, M. H. (2013). Scaffolding and self-scaffolding: Central aspects of development. In *Children's development within social context* (pp. 33–52). Psychology Press.
- * Bilik, Ö., Kankaya, E. A., & Deveci, Z. (2020). Effects of web-based concept mapping education on students' concept mapping and critical thinking skills: A double-blind, randomized, controlled study. *Nurse Education Today*, 86, Article 104312.
- * Bin, H., Zhang, Y., Zhu, Z., Lin, Y., Zhan, Y., & Ma, J. (2022). Research on the influence of collaborative mind mapping strategy in smart classroom on college students' scientific research problem-solving ability and learning anxiety. In *2022 international symposium on educational technology (ISET)* (pp. 23–27). IEEE.
- * Bixler, G. M., Brown, A., Way, D., Ledford, C., & Mahan, J. D. (2015). Collaborative concept mapping and critical thinking in fourth-year medical students. *Clinical Pediatrics*, 54(9), 833–839.
- Bloom, B. (1956). *Taxonomy of educational objectives: Handbook I. Cognitive domain*. New York: McKay.
- Botley, S. P., & Hakim, F. (2014). Argument structure in learner writing: A corpus-based analysis using argument mapping. *Kajian Malaysia*, 32(1), 45–77.
- * Boyadjian-Samawi, Z. (2006). *The effect of concept mapping on critical thinking skills and dispositions of junior and senior baccalaureate nursing students*. Widener University School of Nursing.
- Carey, S., Zaitchik, D., & Bascandziev, I. (2015). Theories of development: In dialog with Jean Piaget. *Developmental Review*, 38, 36–54.
- * Carvalho, D. P. D. S. R. P., Vitor, A. F., Cogo, A. L. P., Bittencourt, G. K. G. D., Santos, V. E. P., & Ferreira, M. A. (2020). Critical thinking in nursing students from two Brazilian regions. *Revista Brasileira de Enfermagem*, 73(1), Article e20170742.
- * Carwie, L. (2009). *The effect of computer-supported argument mapping on the critical thinking skills of undergraduate nursing students*. University of South Alabama.
- * Chen, P. (2020). *The effects of painting performance and creativity on the mind mapping thinking strategy on visual arts curriculum for art class students of junior high school*. Taiwan: National Taiwan Normal University.
- * Chen, S. L., Liang, T., Lee, M. L., & Liao, I. C. (2011). Effects of concept map teaching on students' critical thinking and approach to learning and studying. *Journal of Nursing Education*, 50(8), 466–469.
- * Chen, M. R. A., & Hwang, G. J. (2020). Effects of a concept mapping-based flipped learning approach on EFL students' English speaking performance, critical thinking awareness, and speaking anxiety. *British Journal of Educational Technology*, 51(3), 817–834.
- * Chiu, M. C., & Hwang, G. J. (2024). Enhancing students' critical thinking and creative thinking: An integrated mind mapping and robot-based learning approach. *Education and Information Technologies*, 1–34.
- Chiu, M. M., & Roberts, C. A. (2018). Improved analyses of single cases: Dynamic multilevel analysis. *Developmental Neurorehabilitation*, 21(4), 253–265.
- Compayre, G., & Payne, W. H. (2015). *The history of pedagogy*. Routledge.
- * Crudele, F., & Raffaghelli, J. E. (2023). Promoting critical thinking through argument mapping: A lab for undergraduate students. *Journal of Information Technology Education: Research*, 22, 497–525.
- * Cullen, S., Fan, J., van der Brugge, E., & Elga, A. (2018). Improving analytical reasoning and argument understanding: A quasi-experimental field study of argument visualization. *Npj Science of Learning*, 3(1), 21.
- * D'Antoni, A. V., Zipp, G. P., Olson, V. G., & Cahill, T. F. (2010). Does the mind map learning strategy facilitate information retrieval and critical thinking in medical students? *BMC Medical Education*, 10, 1–11.
- Davies, M. (2011). Concept mapping, mind mapping and argument mapping: What are the differences and do they matter? *Higher Education*, 62, 279–301.
- Dexter, D. D., Park, Y. J., & Hughes, C. A. (2011). A meta-analytic review of graphic organizers and science instruction for adolescents with learning disabilities: Implications for the intermediate and secondary science classroom. *Learning Disabilities Research & Practice*, 26(4), 204–213.
- diSessa, A. A., & Sherin, B. L. (2000). Meta-representation. *The Journal of Mathematical Behavior*, 19(4), 385–398. [https://doi.org/10.1016/S0732-3123\(01\)00051-7](https://doi.org/10.1016/S0732-3123(01)00051-7)
- * Dwyer, C. P., Hogan, M. J., & Stewart, I. (2011). *The promotion of critical thinking skills through argument mapping*.
- * Dwyer, C. P., Hogan, M. J., & Stewart, I. (2012). An evaluation of argument mapping as a method of enhancing critical thinking performance in e-learning environments. *Metacognition and Learning*, 7, 219–244.
- Edwards, S., & Cooper, N. (2010). Mind mapping as a teaching resource. *The Clinical Teacher*, 7(4), 236–239.
- Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *British Medical Journal*, 315, 629–634.
- * Eggert, S., Nitsch, A., Boone, W. J., Nückles, M., & Bögeholz, S. (2017). Supporting students' learning and socioscientific reasoning about climate change—The effect of computer-based concept mapping scaffolds. *Research in Science Education*, 47, 137–159.
- * Elasarag, G. A. A., & Elsabagh, N. E. (2020). The effect of mind mapping on critical thinking skills of undergraduate nursing students. *Pharmacophore*, 11(1), 73–84.
- Eppler, M. J., & Burkhard, R. A. (2008). Knowledge visualization. In *Knowledge management: Concepts, methodologies, tools, and applications* (pp. 781–793). IGI Global.
- * Esfandiari, M. A., Rezvani, E., & Hadian, B. (2021). Impact of critical thinking instruction through argument mapping techniques on Iranian Male and female EFL learners' critical thinking ability and reading skill. *Research in English Language Pedagogy*, 9(1), 70–89.
- * Esquillo, S. (2023). Argument mapping method and learners' critical thinking dispositions in physics. *WVSU Research Journal*, 12(2), 31–46.
- Ezzeldin, S. M. (2017). The effectiveness of using graphic organizers in development of achievement, reduction of cognitive load associated with solving algorithm problems in analytical chemistry and favored learning styles among female secondary school students in Saudi Arabia. *International Journal of Religious Education*, 41(2), 77–124.
- Facione, P. A. (1990). *The California critical thinking skills Test—College level*. Experimental Validation and Content Validity. Technical Report# 1.
- Farrand, P., Hussain, F., & Hennessy, E. (2002). The efficacy of the mind map study technique. *Medical Education*, 36(5), 426–431.
- Fraillon, J., Ainley, J., Schulz, W., Friedman, T., & Gebhardt, E. (2014). *Preparing for life in a digital age: The IEA international computer and information literacy study international report*. Springer Nature.
- Galita, R., & Bonta, E. (2023). Critical thinking at play in language autobiographies. The case of intellectual perseverance. In N. Abid, & E. Bonta's (Eds.), *Exploring the autobiography as a genre and a data collection tool* (pp. 66–86).
- Hansen, B. (2022). *Econometrics*. Princeton University Press.
- Hay, D. B. (2007). Using concept maps to measure deep, surface and non-learning outcomes. *Studies in Higher Education*, 32(1), 39–57.
- * Hazaymeh, W. A., & Alomery, M. K. (2022). The effectiveness of visual mind mapping strategy for improving English language learners' critical thinking skills and reading ability. *European Journal of Educational Research*, 11(1), 141–150.
- * Heglund, B. (2015). *Using computer-assisted argumentation mapping to develop effective argumentation skills in high school advanced placement physics (doctoral dissertation, university of Florida)*.
- Hillmann, P. J. (1993). *Visual literacy in elementary school children: Effects of age, gender and ethnicity on graphic representation and divergent production*. The University of Wisconsin-Madison.
- Hittleman, D. R. (1985). A picture is worth a thousand words... if you know the words. *Childhood Education*, 62(1), 32–36.
- Hoffmann, M. H. (2005). Logical argument mapping: A method for overcoming cognitive problems of conflict management. *International Journal of Conflict Management*, 16(4), 304.
- * Hu, Y., & Hwang, G. J. (2024). Promoting students' higher order thinking in virtual museum contexts: A self-adapted Mobile concept mapping-based problem posing approach. *Education and Information Technologies*, 29(3), 2741–2765.
- * Huang, M. Y., Tu, H. Y., Wang, W. Y., Chen, J. F., Yu, Y. T., & Chou, C. C. (2017). Effects of cooperative learning and concept mapping intervention on critical thinking and basketball skills in elementary school. *Thinking Skills and Creativity*, 23, 207–216.
- * Huang, Y. C., Chen, H. H., Yeh, M. L., & Chung, Y. C. (2012). Case studies combined with or without concept maps improve critical thinking in hospital-based nurses: A randomized-controlled trial. *International Journal of Nursing Studies*, 49(6), 747–754.
- Huedo-Medina, T. B., Sánchez-Meca, J., Marín-Martínez, F., & Botella, J. (2006). Assessing heterogeneity in meta-analysis: Q statistic or I² index? *Psychological Methods*, 11(2), 193–206.
- * Hwang, G. J., Huang, H., Wang, R. X., & Zhu, L. L. (2021). Effects of a concept mapping-based problem-posing approach on students' learning achievements and critical thinking tendency: An application in classical Chinese learning contexts. *British Journal of Educational Technology*, 52(1), 374–493.

- Imsa-ard, P. (2022). Reading better?: Enhancing Thai EFL secondary school students' reading comprehension abilities with the use of graphic organizers. *English Language Teaching*, 15(5), 1–13.
- * Israel, C., Zipp, G. P., D'Abundo, M., & Deluca, D. (2020). Mind mapping to enhance critical thinking skills in physician assistant education: A randomized controlled study. *Journal Of Allied Health*, 49(2), 135–140.
- Kabatas Memiş, E., & Karakuş, E. (2021). An evaluation of academic achievements through the use of argument and concept maps embedded in argumentation based inquiry. *Asia Pacific Education Review*, 22(3), 463–481.
- * Kaddoura, M., Van-Dyke, O., & Yang, Q. (2016). Impact of a concept map teaching approach on nursing students' critical thinking skills. *Nursing and Health Sciences*, 18(3), 350–354.
- * Kamble, S. K., & Tembe, B. L. (2013). The effect of use of concept maps on problem solving performance and attitude in mechanical engineering course. *Procedia - Social and Behavioral Sciences*, 83, 748–754.
- Kansızoğlu, H. B. (2017). The effect of graphic organizers on language teaching and learning areas: A meta-analysis study. *Education and Science*, 42(191), 139–164.
- Kelly, A., Sullivan, M., & Strampel, K. (2023). Generative artificial intelligence: University student awareness, experience, and confidence in use across disciplines. *Journal of University Teaching and Learning Practice*, 20(6), 1–16.
- * Kharismayuni, E., Feronika, T., & Yunita, L. (2021). Implication of thinking maps assisted inquiry model for higher order thinking skills (HOTS) on chemistry. *Journal of Physics: Conference Series*, 1836(1), 12078. IOP Publishing.
- * Khodadady, E., & Ghanizadeh, A. (2011). The impact of concept mapping on EFL learners' critical thinking ability. *English Language Teaching*, 4(4), 49–60.
- * Khrais, H., & Saleh, A. M. (2020). The effect of concept mapping on critical thinking of Jordanian nursing students. *Creative Nursing*, 26(1), 19–24.
- Kmet, L. M., Cook, L. S., & Lee, R. C. (2004). *Standard quality assessment criteria for evaluating primary research papers from a variety of fields*. Krippendorff, K. (2011). *Computing Krippendorff's alpha-reliability*.
- * Laelasari, I., & Anggraeni, S. (2017). Improving critical thinking and metacognition ability using vee diagram through problem-based learning of human respiratory system. In *In international conference on mathematics and science education* (pp. 70–72). Atlantis Press.
- Lakoff, G., & Johnson, M. (2024). *Metaphors we live by*. University of Chicago press.
- Lane, S. (1993). The conceptual framework for the development of a mathematics performance assessment instrument. *Educational Measurement: Issues and Practice*, 12(2), 16–23. <https://doi.org/10.1111/j.1745-3992.1993.tb00529.x>
- Larkin, J. H., & Simon, H. A. (1987). Why a diagram is (sometimes) worth ten thousand words. *Cognitive Science*, 11(1), 65–100.
- * Lazuardini, D. A., Wilujeng, I., & Kuswanto, H. (2019). Optimizing senior high school students' creative thinking skills of optical devices through inductive learning models assisted by e-Mind map. *Journal of Physics: Conference Series*, 1233(1), Article 012029. IOP Publishing.
- Lee, J. (2012). *Development of a visual summarizer design model for digital learning*. Seoul, Korea: Seoul National University. Unpublished doctoral dissertation.
- * Lee, W., Chiang, C. H., Liao, I. C., Lee, M. L., Chen, S. L., & Liang, T. (2013). The longitudinal effect of concept map teaching on critical thinking of nursing students. *Nurse Education Today*, 33(10), 1219–1223.
- Leeds, A. J., Kudrowitz, B., & Kwon, J. (2019). Mapping associations: Exploring divergent thinking through mind mapping. *International Journal of Design Creativity and Innovation*, 7(1–2), 16–29.
- Lim, C. H. (2017). Cultivating higher order thinking dispositions of undergraduate students. *International Journal of Learning and Teaching*, 3, 154–159.
- Lin, L., & Chu, H. (2018). Quantifying publication bias in meta-analysis. *Biometrics*, 74(3), 785–794.
- Lipsey, M. W., & Wilson, D. B. (2001). *Practical meta-analysis*. Sage.
- Liu, Q., Zhong, Z., & Nesbit, J. C. (2024). Argument mapping as a pre-writing activity: Does it promote writing skills of EFL learners? *Education and Information Technologies*, 29(7), 7895–7925.
- Lu, Z., Chiu, M. M., Wang, S., Mao, W., & Lei, H. (2025). Effects of augmented reality on students' higher-order thinking: A meta-analysis. *Journal of Educational Computing Research*, Article 07356331241309644.
- * Maldonado, S. (2014). *Effects of concept mapping on the critical thinking skills of baccalaureate nursing students (doctoral dissertation, walden university)*.
- * Maryam, A., Mohammadreza, D., Abdolhussein, S., Ghobad, R., & Javad, K. (2021). Effect of concept mapping education on critical thinking skills of medical students: A quasi-experimental study. *Ethiopian Journal of Health Sciences*, 31(2).
- Mayer, R. E. (2002). Multimedia learning. In *Psychology of learning and motivation* (Vol. 41, pp. 85–139). Academic Press.
- Meo, G. (2008). Curriculum planning for all learners: Applying universal design for learning (UDL) to a high school reading comprehension program. *Preventing School Failure: Alternative Education for Children and Youth*, 52(2), 21–30.
- * Min, G. S., Albakri, I. S. M. A., Ismail, N., Mokhtar, M. M., Zulkepli, N., Tahir, M. H. M., & Khalid, P. Z. M. (2023). Fostering critical thinking using graphic organizers in English language reading class. *Studies in English Language and Education*, 10(3), 1309–1325.
- * Mohamed El-Sayed, N., Mahmoud Abdel-Azeem, A., & Khaled Abd El-Aziz Zaki, A. (2023). The effect of mind mapping on cognitive achievement and critical thinking skills of nursing students. *International Egyptian Journal of Nursing Sciences and Research*, 3(2), 685–703.
- Morrison, S., Adamson, C., Nibert, A., & Hsia, S. (2004). HESI exams: An overview of reliability and validity. *CIN: Computers, Informatics, Nursing*, 22(4), 220–226.
- * Mutiara, P. A., Achmad, F., Alief, M., Lindsari, L. M., Supriadi, N., Putra, F. G., ... Rahmawati, N. K. (2021). Analysis of mathematical critical thinking skills: The impact of RMS (reading, mind mapping, and sharing) learning model assisted by PPW (props, PowerPoint, and worksheet). *Journal of Physics: Conference Series*, 1796(1), Article 012010. IOP Publishing.
- Nelson, R. R. (2005). *Technology, institutions, and economic growth*. Harvard University Press.
- Nesbit, J. C., & Adesope, O. O. (2006). Learning with concept and knowledge maps: A meta-analysis. *Review of Educational Research*, 76(3), 413–448.
- * Ngajie, B. N., Li, Y., Tiruneh, D. T., & Cheng, M. (2020). Investigating the effects of a systematic and model-based design of computer-supported argument visualization on critical thinking. *Thinking Skills and Creativity*, 38, Article 100742.
- * Nirmala, T., & Shakuntala, B. S. (2011). Concept mapping-An effective tool to promote critical thinking skills among nurses. *Journal of Health and Allied Sciences NU*, 1(4), 21–26.
- Palaniappan, V., Karthikeyan, K., & Mohan, R. (2023). Mind mapping as a novel method in teaching the morphology of skin lesions: A quasi-experimental study. *Journal of Advances in Medical Education & Professionalism*, 11(2), 80.
- Pande, P., & Chandrasekharan, S. (2017). Representational competence: Towards a distributed and embodied cognition account. *Studies in Science Education*, 53(1), 1–43.
- * Pinadito, A., Hayashi, Y., & Hirashima, T. (2023). Promoting students' higher-order thinking with concept mapping recomposition. *IEICE Transactions on Information and Systems*, 106(8), 1262–1274.
- Purnama, Y. I., & Nurdianingsih, F. (2019). The impact of higher order thinking skills (HOTS) instructions in teaching EFL speaking skill from the perspective of students' motivation. *Lingua e Cultura*, 13(4), 313–319.
- Rauscher, W., & Badenhurst, H. (2021). Thinking critically about critical thinking dispositions in technology education. *International Journal of Technology and Design Education*, 31(3), 465–488.
- Redhana, I. W., Widiastari, K., Samsudin, A., & Irwanto, I. (2021). Which is more effective, a mind map or a concept map learning strategy? *Jurnal Cakrawala Pendidikan*, 40(2), 520–531.
- * Ristanto, R., Sabrina, A., & Komala, R. (2022). Critical thinking skills of environmental changes: A biological instruction using guided discovery learning-argument mapping (GDL-AM). *Participatory Educational Research*, 9(1), 173–191.
- Roman, D., Jones, F., Basaraba, D., & Hironaka, S. (2016). Helping students bridge inferences in science texts using graphic organizers. *Journal of Adolescent & Adult Literacy*, 60(2), 121–130.
- Romero García, C., Cazorla, M., & Buzón García, O. (2017). Meaningful learning using concept maps as a learning strategy. *JOTSE: Journal of Technology and Science Education*, 7(3), 313–332.
- Rowland-Bryant, E., Skinner, C. H., Skinner, A. L., Saudargas, R., Robinson, D. H., & Kirk, E. R. (2009). Investigating the interaction of graphic organizers and seductive details: Can a graphic organizer mitigate the seductive-details effect. *Research in the Schools*, 16(2), 29–40.

- Schnotz, W. (2014). An integrated model of text and picture comprehension. In R. E. Mayer (Ed.), *The Cambridge handbook of multimedia learning* (2nd ed., pp. 72–103). Cambridge University Press.
- Schnotz, W., & Bannert, M. (2003). Construction and interference in learning from multiple representation. *Learning and Instruction*, 13(2), 141–156.
- Schwarzer, G., Carpenter, J. R., & Rücker, G. (2015). *Meta-analysis with R*. Springer. <https://doi.org/10.1007/978-3-319-21416-0>
- Scott, D. B. (2011). *Explicit instruction on rhetorical patterns and student-constructed graphic organizers: The impact on sixth-grade students' comprehension of social studies text (doctoral dissertation)*. College Park, Maryland: University of Maryland, The Faculty of the Graduate School.
- Shi, Y., Yang, H., Dou, Y., & Zeng, Y. (2023). Effects of mind mapping-based instruction on student cognitive learning outcomes: A meta-analysis. *Asia Pacific Education Review*, 24(3), 303–317.
- * Silva, H., Lopes, J., Dominguez, C., & Morais, E. (2022). Lecture, cooperative learning, and concept mapping: Any differences on critical and creative thinking development?. *International Journal of Instruction*, 15(1), 765–780.
- * Simper, N., Reeve, R., & Kirby, J. R. (2016). Effects of concept mapping on creativity in photo stories. *Creativity Research Journal*, 28(1), 46–51.
- Smith, V. D., & Darvas, J. W. (2017). Encouraging student autonomy through higher order thinking skills. *Journal of Instructional Research*, 6, 29–34.
- * Su, K. D. (2018). Enhancing students' corresponding reasoning of cognitive performances by animated concept mapping in electrochemistry. *Journal of Baltic Science Education*, 17(4), 662–673.
- * Su, G., & Long, T. (2021). Is the text-based cognitive tool more effective than the concept map on improving the pre-service teachers' argumentation skills?. *Thinking Skills and Creativity*, 41, Article 100862.
- Suardana, I. N., Redhana, I. W., & Yunithasari, N. P. M. (2020). Students' critical thinking skills comparison in discovery learning based on constructing concept mapping and mind mapping. *Journal of Physics: Conference Series*, 1521(4), Article 042089. IOP Publishing.
- * Sun, M., Wang, M., & Wegerif, R. (2019). Using computer-based cognitive mapping to improve students' divergent thinking for creativity development. *British Journal of Educational Technology*, 50(5), 2217–2233.
- Sweller, J., Ayres, P., & Kalyuga, S. (2011). *Cognitive load theory*. Springer.
- Tanujaya, B., Mumu, J., & Margono, G. (2017). The relationship between higher order thinking skills and academic performance of student in mathematics instruction. *International Education Studies*, 10(11), 78–85.
- Thamrin, N. R., & Widodo, P. (2019). Developing higher order thinking skills (HOTS) for reading comprehension enhancement. *Journal of Physics: Conference Series*, 1179(1), Article 012073. IOP Publishing.
- * Tian, Y., Zhang, Y., & Lin, Y. (2023). Effects of concept mapping-based review strategy on students' literature review writing ability and critical thinking tendency. *2023 international symposium on educational technology (ISET)* (pp. 79–83). IEEE.
- Tikhonova, E., & Kudinova, N. (2015). Sophisticated thinking: Higher order thinking skills. *Journal of Language and Education*, 1(3), 12–23.
- * Turkestani, F. A., Zipp, G. P., Al Nufaiei, Z., Al Zhranei, R., Alhadian, F., Hakeem, J., ... Homoud, M. (2024). Mind mapping to enhance critical thinking skills in respiratory therapy education. *Journal of Education and Health Promotion*, 13(1), 198.
- * Twyman, T. (2003). *Effects of a conceptually framed, problem/solution/effect graphic organizer on content comprehension and problem-solving skills for seventh-grade social studies students*. Doctoral dissertation, University of Oregon.
- * Uçar, B., & Demiraslan Çevik, Y. (2020). The effect of argument mapping supported with peer feedback on pre-service teachers' argumentation skills. *Journal of Digital Learning in Teacher Education*, 37(1), 6–29.
- Uttal, D. H., & Doherty, K. O'. (2008). Comprehending and learning from 'visualizations': A developmental perspective. *Visualization: Theory and practice in science education* (pp. 53–72). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-1-4020-5267-5_3
- * Van Boxtel, C., & Veerman, A. (2001). Diagram-mediated collaborative learning: Diagrams as tools to provoke and support elaboration and argumentation. In *Proceedings of the first European conference on computer-supported collaborative learning (EuroCSCL2001)*, maastricht, the Netherlands (pp. 22–24).
- Vøllestad, J., Nielsen, M. B., & Nielsen, G. H. (2012). Mindfulness and acceptance-based interventions for anxiety disorders: A systematic review and meta-analysis. *British Journal of Clinical Psychology*, 51(3), 239–260.
- * Wang, Y. H., & Liao, H. C. (2014). Promoting English oral communication and higher-order thinking in Taiwanese ESL students through the use of knowledge visualization techniques. *Perceptual and Motor Skills*, 118(3), 691–708.
- Watson, M. K., Pelkey, J., Noyes, C. R., & Rodgers, M. O. (2016). Assessing conceptual knowledge using three concept map scoring methods. *Journal of Engineering Education*, 105(1), 118–146.
- * Wei, L., Firetto, C. M., Murphy, P. K., Li, M., Greene, J. A., & Croninger, R. M. (2019). Facilitating fourth-grade students' written argumentation: The use of an argumentation graphic organizer. *The Journal of Educational Research*, 112(5), 627–639.
- * Wheeler, L. A., & Collins, S. K. (2003). The influence of concept mapping on critical thinking in baccalaureate nursing students. *Journal of Professional Nursing*, 19(6), 339–346.
- * Widiana, I. W., & Jampel, I. N. (2016). Improving students' creative thinking and achievement through the implementation of multiple intelligence approach with mind mapping. *International Journal of Evaluation and Research in Education*, 5(3), 246–254.
- * Wu, B., Wang, M., Grotzer, T. A., Liu, J., & Johnson, J. M. (2016). Visualizing complex processes using a cognitive-mapping tool to support the learning of clinical reasoning. *BMC Medical Education*, 16, 1–8.
- * Wu, H. Z., & Wu, Q. T. (2020). Impact of mind mapping on the critical thinking ability of clinical nursing students and teaching application. *Journal of International Medical Research*, 48(3), Article 0300060519893225.
- * Yang, H., & Yin, S. (2015). Applying graphic reasoning simulation tool to explore the scientific creativity. In *In international conference on computers in education*.
- * Yilmaz-Na, E., & Sönmez, E. (2023). Unfolding the potential of computer-assisted argument mapping practices for promoting self-regulation of learning and problem-solving skills of pre-service teachers. *Computers & Education*, 193, Article 104683.
- Zhan, Y., Yan, Z., Wan, Z. H., Wang, X., Zeng, Y., Yang, M., & Yang, L. (2023). Effects of online peer assessment on higher-order thinking: A meta-analysis. *British Journal of Educational Technology*, 54(4), 817–835.
- Zohar, A., & Ben David, A. (2009). Paving a clear path in a thick forest: A conceptual analysis of a metacognitive component. *Metacognition and Learning*, 4(3), 177–195.